Mandava Sukesh

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Education

Amrita Vishwa Vidyapeetham **October 2021 – May 2025**
Bachelor of Technology in Artificial Intelligence (First class with distinction) *Chennai, India*
CGPA - 8.91

Experience

SG Analytics **January 2025 – July 2025**
Data Science Intern *Pune, Maharashtra*

- Engineered a smart web crawler and data pipeline using Python, Scrapy, FastAPI, and AWS S3, automating extraction of 500+ URLs/minute with 95% accuracy and visualized via an interactive Streamlit dashboard.
- Automated SFDR-optimized CIM system that extracts data and KPIs from unstructured corporate documents, into Excel outputs, enhancing workflows and reporting efficiency, for streamlined financial analysis.
- Built a smart SWOT Analysis System using Streamlit UI, RAG, Amazon Bedrock, integrating Annual, SEC 10-K reports and web data, automating report generation with 95% factual summarization, reducing manual research.
- Integrated OpenWebUI with Amazon Bedrock to create private document interaction without public exposure.

Projects

LangGraph Agentic AI: Building Multi-LLM Chatbot with Tool Integration **May 2025**

∗ Leveraged Langgraph to orchestrate a multi-LLM chatbot, integrating various LLM providers and routing to tools.
∗ Integrated tools (Tavily, Arxiv, Wikipedia) in chatbot, demonstrating 40% faster tool integration vs traditional methods.
∗ Designed a Streamlit UI for chatbot, featuring advanced display options,dual modes,scalable for 300+ concurrent users.

AI Powered financial research assistant for equity analysis **March 2025**

∗ Developed an LLM-powered news research tool for equity analysis, enabling answer extraction from 3 news URLs.
∗ Engineered content processing with Langchain and integrated OpenAI's models for intelligent information retrieval.
∗ Built a Streamlit interface for seamless URL and natural language query input, showcasing Gen AI in finance.

Enhancing Super-Resolution: Comparative Study of SRGAN and ESRGAN for SISR **December 2024**

∗ Explored and compared SRGAN and ESRGAN for single-image super-resolution,contributing to enhanced image quality.
∗ Optimized ESRGAN,incorporating RRDB resulting in SSIM up to 0.962, recovering finer texture details.
∗ Trained GANs on large datasets including DIV2K (1000 HR photos) and Flickr2K producing more realistic images.

Technical Skills

Languages: Java, Python
Developer Tools: NetBeans, pycharm, Jupyter Notebook, VS Code, Cursor, AWS SageMaker, n8n, Zapier
Frameworks: FastAPI, Streamlit, TensorFlow, Langchain, Langgraph, CrewAI
Databases: AWS S3, MySQL, MongoDB

Leadership / Extracurricular

Neuronix Club **August 2023 – Present**
Domain head of Neuralearn domain *Amrita Vishwa Vidyapeetham*

∗ Led a team of 4 students driving initiatives to enhance knowledge and practical skills in ML and DL.

Tantrotsav-23 & 24 **April 2023 & March 2024**
Resource person for XGen Learning *Amrita Vishwa Vidyapeetham*

∗ Conducted a workshop, providing hands-on sessions focused on various concepts of ML in 2 sessions for 50+ students.
∗ Coordinated and managed technical **(Codeblind)** with 30+ participating teams and non-technical **(Squid Game)** with 20+ participating teams.

Smart India Hackathon **August 2022**
Finalist and Runner-up *KLE Technological University*

∗ Achieved "Finalist and Runner-up" recognition in the Smart India Hackathon by developing an Android application for a Government Assets Management System within 2 days.

Publications

ICAECC 2023 **June 2024**

∗ **Paper:** Stroke Prediction Using Machine Learning LINK

InC4 2024 **September 2024**

· **Paper:** Detecting AI-generated images with CNN and Interpretation using Explainable AI LINK